Experiment 1: Sionna RT Channel Impulse Response at 5.9 GHz

1. Objective

To investigate radio-wave propagation in a simulated street-canyon environment at 5.9 GHz using NVIDIA Sionna RT. The experiment aims to identify multipath propagation delays and relative path magnitudes as a baseline for future V2X Integrated Sensing and Communication (ISAC) experiments.

2. Experimental Setup

- **Simulation tool:** NVIDIA Sionna RT
- **Execution backend:** Mitsuba 3 with LLVM CPU backend
- **Carrier frequency:** 5.9 GHz
- **Environment:** Built-in `simple_street_canyon` scene
- **Transmitter position:** (-20, 0, 2) m
- **Receiver position:** (20, 0, 2) m
- **Transmitter–receiver separation:** 40 m
- **Antenna configuration:** Single-element isotropic antenna arrays
- **Ray-tracing solver:** Sionna RT `PathSolver`
- **Number of valid path entries:** 12

The simulation was executed successfully without requiring the OptiX backend.

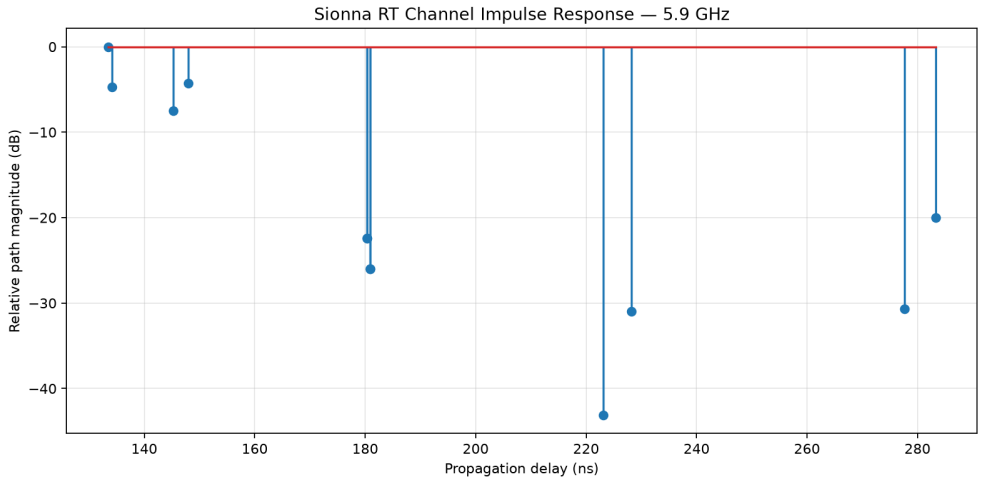
3. Graph Description

The generated graph, `cir_5p9GHz.png`, displays relative propagation-path magnitude against propagation delay.

- **X-axis:** Propagation delay in nanoseconds (ns).
- **Y-axis:** Relative path magnitude in decibels (dB).
- **Blue stems:** Individual valid propagation-path entries.
- **0 dB reference:** Strongest path, used for magnitude normalization.

The plotted quantity is the relative magnitude of individual path coefficients at their corresponding delays. It is not yet a sampled, bandwidth-limited channel impulse response.

4. Numerical Results



- **Minimum reported delay:** 133.426 ns
- **Maximum reported delay:** 283.282 ns
- **Delay spread across reported entries:** Approximately 149.856 ns
- **Shortest-delay equivalent propagation distance:** Approximately 40.0 m
- **Longest-delay equivalent propagation distance:** Approximately 84.9 m
- **Valid path entries:** 12 out of 12

The equivalent propagation distance was calculated using:

$d = c\tau$

where:

- d is the total propagation distance along a path;
- c is the speed of light, approximately 3×10^8 m/s;
- τ is the propagation delay.

The equivalent distance is not necessarily the direct geometric distance between the transmitter and receiver.

5. Interpretation and Findings

1. **Multipath propagation:** The simulation produces multiple valid propagation-path entries with different delays and relative magnitudes.

2. **Strongest path:** The earliest reported path has a delay of approximately 133.426 ns, corresponding to an equivalent distance of approximately 40 m. This is consistent with the transmitter–receiver separation and may represent the direct path. Its identity should be verified using path interactions and geometry.
3. **Weaker propagation components:** Several paths have substantially lower magnitudes than the strongest path. These may represent reflected or other indirect paths, but their physical origins require verification.
4. **Delay variation:** The reported propagation delays span approximately 150 ns. Such differences can contribute to frequency-selective fading and affect communication and sensing performance.
5. **Complex path coefficients:** Sionna RT exposes the path coefficients as two real-valued tensors in the installed software version. These were interpreted as the real and imaginary components for the relative-magnitude calculation.
6. **Simulation validation:** Successful scene loading, path solving, and graph generation confirm that the basic ray-tracing workflow is operational on the LLVM CPU backend.

6. Limitations

- The scene is a built-in static street-canyon environment, not yet a custom V2X road scene containing explicitly modelled vehicles.
- The current graph does not independently establish which paths correspond to direct propagation, building reflections, or vehicle reflections.
- The plotted response is a path-magnitude representation rather than a sampled OFDM channel impulse response.
- No moving objects, Doppler processing, target detection, or range estimation have been implemented.
- No measurements have yet been collected using the USRP B210 hardware.
- The results are simulation outputs and should not be presented as experimental hardware measurements.

7. Next Tasks

Task 1 — Validate the propagation paths

- Inspect path validity and interaction data.
- Identify the direct path and classify reflected paths.
- Compare the direct-path delay with the expected 40 m geometric distance.

Task 2 — Improve channel analysis

- Calculate the complex path magnitudes and phases.
- Generate a properly defined channel impulse response.
- Investigate delay spread and frequency-selective channel behaviour.

Task 3 — Implement OFDM waveform processing

- Define OFDM parameters, including bandwidth, subcarrier spacing, FFT size, and cyclic prefix.
- Generate a known OFDM waveform.
- Apply the simulated channel and estimate the channel response.
- Produce a range profile and evaluate range resolution.

Task 4 — Introduce sensing targets

- Create a custom road scene with a vehicle or other target.
- Isolate the target's reflected contribution.
- Investigate target-range estimation and, subsequently, Doppler estimation.

Task 5 — Prepare hardware validation

- Establish a repeatable simulation baseline.
- Configure the two USRP B210 devices for a controlled 5.9 GHz experiment.
- Compare simulated and measured channel or range-profile results, accounting for hardware calibration, synchronization, antennas, and regulatory requirements.

8. Current Status

Completed: CPU/LLVM backend validation, street-scene loading, transmitter/receiver configuration, ray-tracing execution, path-delay extraction, and relative path-magnitude plotting.

Next milestone: Validate the physical origin of the simulated paths and implement an OFDM-based range-estimation pipeline.

9. Overall Conclusion

The experiment establishes a working Sionna RT propagation simulation at 5.9 GHz and demonstrates multiple propagation-path contributions with different delays and relative magnitudes. It provides a baseline for developing OFDM-based V2X-ISAC sensing experiments. Further work is required to validate path identities, model explicit sensing targets, estimate target range from a waveform, and compare simulation results against SDR measurements.